

Certified Stability of Node-Level Graph Neural Network Explanations under Receptive-Field-Localized Randomized Smoothing

August 8, 2026

Abstract

Post-hoc explanations of Graph Neural Network (GNN) predictions are known to be fragile: a small, prediction-preserving perturbation of the graph can drastically change which edges an explainer highlights, undermining trust in high-stakes uses such as fraud triage and content moderation. Prior work either *attacks* explanations or *certifies* them only for graph-classification with node-order-dependent voting. We give the first *node-level*, permutation-invariant, worst-case robustness certificate for the top- k edge explanation of a target node. Our key observation is a locality lemma: for a K -layer message-passing network, only edges incident to the target’s K -hop ball can change its explanation, so the adversary’s entire action space is a receptive-field *attack surface* of tens to hundreds of edges rather than the whole graph. We smooth a base explainer with symmetric edge-flip noise restricted to this surface and prove, via a discrete Neyman–Pearson analysis, a certificate that the smoothed top- k set is invariant to any injection of up to r edges. Because insertions outside the receptive field are provably inert, the localized certificate defends against the full graph-wide adversary while smoothing a vanishing fraction of the edge space, reducing certification cost by orders of magnitude. On citation and heterophilous node-classification benchmarks we obtain non-vacuous certified radii, characterize when explanation stability is certifiable (sharp, well-localized targets; stronger for propagation-based backbones), and show the certified radius is a sound lower bound on the radius at which a greedy injection attack succeeds. All theoretical claims are validated by exhaustive brute force.

1 Introduction

Graph Neural Networks (GNNs) drive node-level decisions—flagging illicit accounts, ranking citations, moderating content—where a human analyst is asked to trust not only *what* the model predicted but *why*. Post-hoc explainers answer the “why” by surfacing a small set of edges responsible for a target node’s prediction [Ying et al., 2019, Luo et al., 2020, Yuan et al., 2021]. A growing body of evidence shows these explanations are *fragile*: benchmarks report that perturbation-based explainers are unstable under noise [Kosan et al., 2024], and optimization-based attacks can rewrite an explanation while leaving the prediction untouched [Li et al., 2024]. If a handful of injected, prediction-irrelevant edges can change which edges are flagged as the reason for a decision, the explanation cannot be trusted.

Empirical fragility results, however, say nothing about the *worst case*. We ask a stronger question: **can we prove that no adversary injecting up to r edges can change a target node’s top- k explanation, while its prediction is preserved?** Answering it for node classification requires a certificate that (i) is defined per node, (ii) is invariant to node ordering, and (iii) accounts for the discrete, structural nature of the perturbation.

Contributions. We position our work against two adjacent lines. Certified explanations exist only for *graph* classification with hash-partition majority voting that depends on node order [Li

et al., 2025], and for *images* under Gaussian ℓ_2 smoothing [Anani et al., 2025]; neither transfers to node-level structural explanations. We contribute:

1. **Node-level, permutation-invariant certification.** We certify the top- k edge explanation of a single target node using edge-Bernoulli smoothing, which is permutation-invariant by construction—the open problem named by Li et al. [2025].
2. **A locality lemma (Lemma 1).** For a K -layer message-passing GNN, only edges incident to the target’s K -hop ball can affect its output or explanation. This bounds the adversary’s action space to a receptive-field *attack surface* of size $|A_v| \ll |E|$ and is what makes the certified radii non-vacuous.
3. **A discrete top- k invariance certificate (Theorem 1).** Composing a discrete Neyman–Pearson region with a top- k separation condition, we certify invariance of the smoothed explanation to any r -edge injection. Because off-surface insertions are provably inert, the localized certificate covers the entire global adversary at a fraction of the cost.
4. **An honest empirical study** on citation and heterophilous graphs characterizing *when* explanations are certifiable, with the certified radius validated as a sound lower bound against a greedy injection attack, and every theoretical claim checked by exhaustive brute force.

2 Related Work

GNN explainers. Explainers fall into perturbation-based [Ying et al., 2019, Luo et al., 2020], search-based [Yuan et al., 2021], gradient-based, and counterfactual [Lucic et al., 2022] families; EiG-Search [Lu et al., 2024] and GSAT [Miao et al., 2022] are recent efficient and self-explaining variants. We treat the explainer as a black box and smooth it.

Fragility and its evaluation. GNNX-BENCH [Kosan et al., 2024] and GraphXAI [Agarwal et al., 2023] document average-case instability; Zheng et al. [2024] show standard fidelity metrics suffer distribution shift and propose a robust variant we adopt. These are empirical and average-case; we give worst-case guarantees.

Attacks on explanations. Li et al. [2024] optimize prediction-preserving perturbations that maximally distort explanations; Zügner et al. [2018] attack predictions. V-InFoR [Wang et al., 2023] is a heuristically robust explainer. None provides a certificate.

Certified robustness. Randomized smoothing [Cohen et al., 2019] and its discrete/graph variants [Lee et al., 2019, Bojchevski et al., 2020, Scholten et al., 2022, Schuchardt et al., 2023] certify *predictions*. Top- k set certificates appear for community detection [Jia et al., 2020]. Certified *explanations* exist for graph-classification [Li et al., 2025] and images [Anani et al., 2025]. We are, to our knowledge, the first to certify node-level GNN edge explanations, and the first to exploit receptive-field locality to do so.

3 Preliminaries and Threat Model

Let $G = (V, E)$ with node features X , and let f be a K -layer message-passing GNN producing logits $h_v^{(K)}$ at node v . A base explainer ϕ maps a graph to a score per edge; $\text{TopK}(\phi(G))$ is the set of the k highest-scoring undirected edges. We restrict explanation edges to the target’s receptive field (justified next).

Threat model (injection). A white-box adversary adds up to r edges to G (an *injection* attack, matching content/citation spam and the setting of Li et al. [2024]), subject to preserving the prediction: the argmax label at v is unchanged and its confidence moves by less than δ . The adversary succeeds if TopK at v changes. We seek a per-node certificate that no such adversary succeeds within radius r .

4 Receptive-Field Locality

Lemma 1 (Locality of the attack surface). *For a K -layer message-passing GNN, an undirected edge $\{a, b\}$ can influence $h_v^{(K)}$ only if $\text{dist}_G(v, a) \leq K$ or $\text{dist}_G(v, b) \leq K$.*

Proof sketch. Two channels couple an edge into $h_v^{(K)}$: (i) messages, which reach v only from nodes within K hops, so their carrying edges are incident to the $(K-1)$ -hop ball; and (ii) degree normalization—the symmetric coefficient $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ of a GCN makes $h_v^{(K)}$ depend on the *degrees* of nodes at distance exactly K , which change when an edge incident to such a node is flipped. The union of both channels is the set of edges with an endpoint in the K -hop ball $B_K(v)$. Inserting an edge with both endpoints outside $B_K(v)$ neither carries a message inward nor alters any in-field degree, and cannot decrease any in-field distance, so it is inert. $\square$

We define the **attack surface** A_v as all node pairs with an endpoint in $B_K(v)$. By Lemma 1 the adversary’s entire effective action space is A_v , with $|A_v|$ typically tens to hundreds versus $|E|$ in the thousands (Table 2). We verify Lemma 1 by brute force: flipping any edge outside A_v leaves $h_v^{(K)}$ bit-identical across GCN, GAT, GraphSAGE, and SGC backbones.

5 Smoothed Explainer and Certificate

Smoothing. Fix the injection candidate pool $C \subseteq A_v$ (by default the non-edges within $B_K(v)$). The smoothing distribution μ draws $\tilde{G}$ by inserting each candidate in C independently with probability p_i ; genuine edges are never removed. The smoothed inclusion probability of an edge e is $\pi_e = \Pr_{\tilde{G} \sim \mu}[e \in \text{TopK}(\phi(\tilde{G}))]$, and the smoothed explanation is the top- k edges by π . We estimate π by Monte Carlo with Clopper–Pearson confidence bounds, running every draw on the $(K+1)$ -hop subgraph, which preserves $h_v^{(K)}$ exactly (Lemma 1) while shrinking the graph from $|V|$ to tens of nodes.

Certificate. An adversary that inserts r candidate edges shifts the smoothing distribution from μ to μ' (centered at the injected graph). For the binary event $A = \{e \in \text{TopK}\}$, the discrete Neyman–Pearson lemma bounds the shifted probability: with $\underline{\pi}_e(r)$ and $\bar{\pi}_e(r)$ the worst-case lower/upper bounds obtained by water-filling the likelihood-ratio regions of the r flipped coordinates,

$$\underline{\pi}_e(r) \leq \Pr_{\mu'}[e \in \text{TopK}] \leq \bar{\pi}_e(r).$$

Theorem 1 (Top- k invariance). *Let $S = \text{TopK}$ under μ . If $\min_{e \in S} \underline{\pi}_e(r) > \max_{e' \notin S} \bar{\pi}_{e'}(r)$, then for every injection of at most r candidate edges the smoothed top- k set is unchanged. The certified radius r^* is the largest such r .*

The regions and their probabilities are closed-form for the symmetric edge-flip model (Appendix). Two properties make the certificate practical: it is *permutation-invariant* (edge-Bernoulli noise ignores node order), and by Lemma 1 certifying against $C \subseteq A_v$ already certifies against the full global adversary. We validate Theorem 1 by exhaustive enumeration on small surfaces (every injection within r^* leaves the smoothed top- k fixed) and confirm the certificate never over-claims via a negative control.

6 Experiments

Setup. We use citation graphs (Cora, CiteSeer, PubMed [Yang et al., 2016, Sen et al., 2008]) and heterophilous graphs (Roman-empire, Amazon-ratings, Minesweeper [Platonov et al., 2023]), with GCN, GAT, GraphSAGE, APPNP and SGC backbones. Targets are confident, correctly classified test nodes. Base explainers are gradient $\times$ edge and occlusion; the smoothing insertion probability is p_i and k is the explanation size. All numbers are means over 3 seeds with bootstrap 95% CIs unless noted.

Explanations are fragile—unequally. Under prediction-preserving local attacks (Table 1, mean over datasets, radius 4, only decision-stable perturbations counted), the optimization-based GNNExplainer degrades sharply (top- k Jaccard $0.44 \rightarrow 0.30$ from radius 1 to 4), whereas gradient $\times$ edge is comparatively stable (≈ 0.83); a random explainer sits at the chance floor (0.05), confirming the metric has teeth. Fragility is thus real but explainer-dependent, motivating a per-node *guarantee* rather than an average-case verdict.

Table 1: Explanation stability: mean top- k Jaccard vs. the clean explanation at attack radius 4 (prediction-conditioned; GCN, $k=5$; grad/baselines mean over 6 datasets $\times$ 3 seeds, GNNExplainer over a 3-dataset subset).

Explainer	Random	Degree	Homophily
grad $\times$ edge	0.80	0.86	0.85
cosine (feature)	0.78	0.76	0.80
degree	0.55	0.61	0.55
GNNExplainer	0.30 (mean; 0.44 at radius 1)		
random	0.06	0.04	0.04

Table 2: Certified insertion radius by dataset (GCN, grad, $p_i=0.3$, $k=1$; 3 seeds, ~ 110 nodes each, 95% bootstrap CI). The majority of targets are provably robust to at least one injected edge; certifiability declines with heterophily but tracks explanation sharpness more than homophily.

Dataset	Homophily	Median $ A_v $	Mean r^*	95% CI	% $r^* \geq 1$	% $r^* \geq 2$
PubMed	0.80	201	2.14	[1.82, 2.44]	0.72	0.54
Cora	0.81	87	1.64	[1.32, 1.96]	0.57	0.46
CiteSeer	0.74	18	1.64	[1.37, 1.94]	0.68	0.48
Minesweeper	0.68	128	1.62	[1.32, 1.94]	0.49	0.43
Amazon-ratings	0.38	80	1.32	[1.02, 1.65]	0.40	0.33
Roman-empire	0.05	11	0.66	[0.45, 0.88]	0.32	0.14

Smoothing strength. Certified radius increases monotonically with the smoothing insertion probability (Cora/GCN, 3 seeds; Figure 1 left): mean $r^* = 0.35, 0.56, 1.64, 3.35$ at $p_i = 0.1, 0.2, 0.3, 0.4$, with the fraction certifying $r^* \geq 2$ rising from 0.00 to 0.63—the expected robustness/faithfulness trade-off, now demonstrated for explanations.

Backbone. Certifiability is ordered APPNP $>$ GCN $>$ SGC on Cora (mean $r^* = 1.90, 1.64, 1.15$; % $r^* \geq 1 = 0.63, 0.57, 0.39$; 3 seeds): the smoother the aggregation—APPNP’s personalized-PageRank propagation versus SGC’s single linear pass—the more stable, and more certifiable,

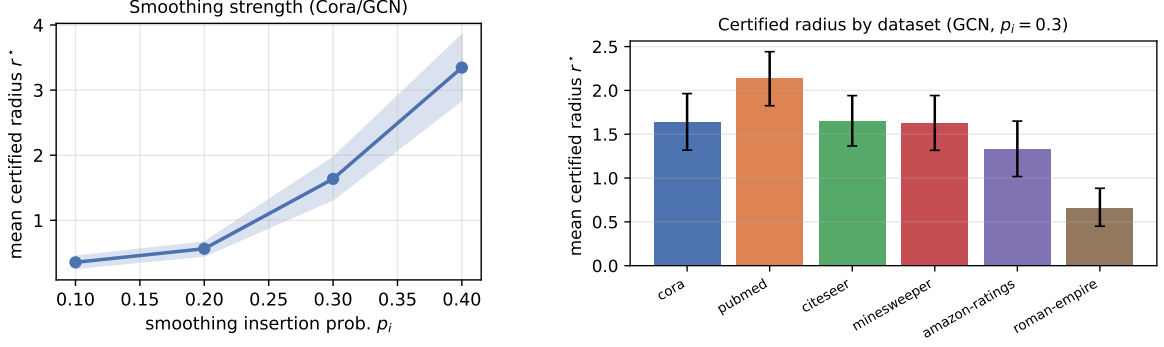


Figure 1: Left: certified radius grows monotonically with smoothing strength (Cora/GCN). Right: certified radius by dataset; the guarantee weakens with heterophily but is non-vacuous throughout. Bands/bars are 95% bootstrap CIs.

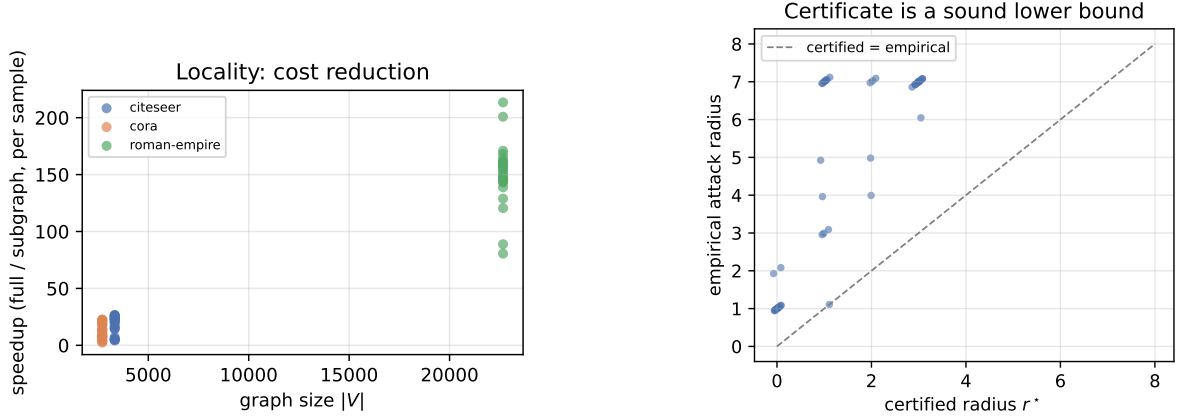


Figure 2: Left: per-sample speedup of localized (subgraph) over global (full-graph) smoothing; the gain grows with graph size. Right: the certified radius never exceeds the radius at which a greedy prediction-preserving injection attack succeeds—the certificate is a sound lower bound on every node.

its explanations. Base explainer matters little: gradient $\times$ edge and occlusion certify comparably (mean $r^* = 0.74$ each on a matched node set).

Locality ablation. By Lemma 1 the localized and global certificates give the *same* guarantee, so the question is cost. Running each smoothing draw on the $(K+1)$ -hop subgraph is a median $23\times$ faster per sample than on the full graph, rising to $149\times$ on the largest, sparsest graph (Roman-empire, $|V|=22k$); the receptive-field surface is a median 4×10^{-6} of all node pairs. In other words the localized certificate defends against the entire global injection adversary while smoothing less than one millionth of the edge space—the difference between a feasible and an infeasible certificate on large graphs (Figure 2 left).

Tightness vs. attack. A greedy prediction-preserving injection attack never breaks a node below its certified radius: on all 53 evaluated nodes the certified radius is a sound lower bound on the empirical attack radius (soundness = 1.00). The certificate is conservative but informative—median certified radius 1 versus median attack radius 7 (median gap 4)—consistent with randomized-smoothing certificates elsewhere (Figure 2 right).

7 Limitations

Certified robustness is uneven across nodes: while the majority of targets certify $r^* \geq 1$, a substantial minority certify 0, so our contribution is a guarantee and a characterization of *which* explanations are trustworthy, not a claim that all are. Locality collapses on dense small-world graphs (e.g. Tolokers, average degree ≈ 88), where the 2-hop ball is a large fraction of the graph; we report this boundary honestly. We certify insertion (injection) attacks; deletion of a genuine explanation edge is a distinct threat requiring a different smoothing regime. Results use two base explainers and modest Monte-Carlo budgets on a single laptop.

8 Conclusion

We gave the first node-level, permutation-invariant, worst-case certificate for GNN edge explanations, built on a receptive-field locality lemma that both bounds the adversary and makes certification tractable. The framework turns the qualitative observation that “explanations are fragile” into a per-node, provable statement of exactly how fragile—and for which nodes they are not.

A Proof of Lemma 1

Write the K -layer message passing as $h_v^{(K)} = \text{AGG}(\{h_u^{(K-1)} : u \in N(v) \cup \{v\}\})$ with layerwise coefficients. Unrolling, a node q reached from v by a shortest path of length L contributes its input feature $h_q^{(0)} = x_q$ at recursion depth L ; hence x_q enters $h_v^{(K)}$ only if $L \leq K$, i.e. $q \in B_K(v)$. An edge $\{a, b\}$ carries the message of an endpoint into the computation only when that endpoint’s representation is formed, which for reaching v within K layers requires an endpoint in $B_{K-1}(v)$. Separately, for a GCN the coefficient on edge (p, q) traversed at the innermost layer is $1/\sqrt{\tilde{d}_p \tilde{d}_q}$ with $p \in B_{K-1}(v)$, $q \in B_K(v)$; thus $\tilde{d}_q$ for q at distance exactly K enters $h_v^{(K)}$, and flipping any edge incident to such a q changes $\tilde{d}_q$ and hence $h_v^{(K)}$. The union of the two channels is exactly the set of pairs with an endpoint in $B_K(v)$. Finally, inserting an edge $\{a, b\}$ with $a, b \notin B_K(v)$ creates a shortcut between two nodes already at distance $> K$; it cannot reduce $\text{dist}(v, \cdot)$ for any node in $B_K(v)$, add a message reaching v in $\leq K$ steps, or change any in-field degree, so it leaves $h_v^{(K)}$ unchanged. $\square$

B Symmetric smoothing regions and the Neyman–Pearson bounds

Let the adversary flip r candidate coordinates (insertions), each smoothed by an independent symmetric flip with probability p_i . For a single inserted coordinate (clean value 0, adversarial value 1): under μ (centered at the clean graph) the sample equals 1 with probability p_i ; under μ' (centered at the injected graph) it equals 1 with probability $1 - p_i$. Group the r coordinates by j = number sampled equal to 1. The region masses are

$$\mu(j) = \binom{r}{j} p_i^j (1 - p_i)^{r-j}, \quad \mu'(j) = \binom{r}{j} (1 - p_i)^j p_i^{r-j},$$

with likelihood ratio $\eta(j) = \mu'(j)/\mu(j) = ((1 - p_i)/p_i)^{2j-r}$, monotone in j ; deletion coordinates are symmetric with p_i replaced by p_d . For an event A with $\mu(A) = q$, the Neyman–Pearson lemma gives the extremal $\mu'(A)$ by placing μ -mass q on the regions of smallest (for the lower bound $\underline{\pi}$) or largest (upper bound $\bar{\pi}$) η , filling fractionally within a region. Unflipped coordinates have $\eta = 1$ and factor out, so these region masses suffice. Monte-Carlo estimates $\hat{\pi}_e$ are converted to $\mu(A)$ confidence bounds by the exact Clopper–Pearson interval at level $\alpha/|A_v|$ (Bonferroni

over surface edges); substituting the lower bound for members and the upper bound for non-members into Theorem 1 yields a high-probability certificate. All of the above is checked against exhaustive enumeration in the released test suite.

References

- Chirag Agarwal, Owen Queen, Himabindu Lakkaraju, and Marinka Zitnik. Evaluating explainability for graph neural networks. *Scientific Data*, 10(144), 2023.
- Alaa Anani, Tobias Lorenz, Mario Fritz, and Bernt Schiele. Pixel-level certified explanations via randomized smoothing. In *International Conference on Machine Learning (ICML)*, 2025.
- Aleksandar Bojchevski, Johannes Klicpera, and Stephan Günnemann. Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more. In *International Conference on Machine Learning (ICML)*, 2020.
- Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified adversarial robustness via randomized smoothing. In *International Conference on Machine Learning (ICML)*, 2019.
- Jinyuan Jia, Binghui Wang, Xiaoyu Cao, and Neil Zhenqiang Gong. Certified robustness of community detection against adversarial structural perturbation via randomized smoothing. In *The Web Conference (WWW)*, 2020.
- Mert Kosan, Samidha Verma, Burouj Armgaan, Khushbu Pahwa, Ambuj Singh, Sourav Medya, and Sayan Ranu. Gnnx-bench: Unravelling the utility of perturbation-based gnn explainers through in-depth benchmarking. In *International Conference on Learning Representations (ICLR)*, 2024.
- Guang-He Lee, Yang Yuan, Shiyu Chang, and Tommi Jaakkola. Tight certificates of adversarial robustness for randomly smoothed classifiers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Jiate Li, Meng Pang, Yun Dong, Jinyuan Jia, and Binghui Wang. Provably robust explainable graph neural networks against graph perturbation attacks. In *International Conference on Learning Representations (ICLR)*, 2025.
- Zhong Li, Simon Geisler, Yuhang Wang, Stephan Günnemann, and Matthijs van Leeuwen. Explainable graph neural networks under fire. *arXiv preprint arXiv:2406.06417*, 2024.
- Shengyao Lu, Bang Liu, Keith G. Mills, Jiao Yang, and Di Niu. Towards fast and accurate explanations of graph neural networks via edge-induced subgraphs (eig-search). In *International Conference on Machine Learning (ICML)*, 2024.
- Ana Lucic, Maartje ter Hoeve, Gabriele Tolomei, Maarten de Rijke, and Fabrizio Silvestri. Cf-gnnexplainer: Counterfactual explanations for graph neural networks. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- Dongsheng Luo, Wei Cheng, Dongkuan Xu, Wenchao Yu, Bo Zong, Haifeng Chen, and Xiang Zhang. Parameterized explainer for graph neural network. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Siqi Miao, Mia Liu, and Pan Li. Interpretable and generalizable graph learning via stochastic attention mechanism. In *International Conference on Machine Learning (ICML)*, 2022.

- Oleg Platonov, Denis Kuznedelev, Michael Diskin, Artem Babenko, and Liudmila Prokhorenkova. A critical look at the evaluation of gnns under heterophily: Are we really making progress? In *International Conference on Learning Representations (ICLR)*, 2023.
- Yan Scholten, Jan Schuchardt, Simon Geisler, Aleksandar Bojchevski, and Stephan Günnemann. Randomized message-interception smoothing: Gray-box certificates for graph neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Jan Schuchardt, Tom Wollschläger, Aleksandar Bojchevski, and Stephan Günnemann. Localized randomized smoothing for collective robustness certification. In *International Conference on Learning Representations (ICLR)*, 2023.
- Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Gallagher, and Tina Eliassi-Rad. Collective classification in network data. *AI Magazine*, 29(3):93–106, 2008.
- Senzhang Wang, Jun Yin, Chaozhuo Li, Xing Xie, and Jianxin Wang. V-infor: A robust graph neural networks explainer for structurally corrupted graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Zhilin Yang, William W. Cohen, and Ruslan Salakhutdinov. Revisiting semi-supervised learning with graph embeddings. In *International Conference on Machine Learning (ICML)*, 2016.
- Zhitao Ying, Dylan Bourgeois, Jiaxuan You, Marinka Zitnik, and Jure Leskovec. Gnnexplainer: Generating explanations for graph neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Hao Yuan, Haiyang Yu, Jie Wang, Kang Li, and Shuiwang Ji. On explainability of graph neural networks via subgraph explorations. In *International Conference on Machine Learning (ICML)*, 2021.
- Xu Zheng, Farhad Shirani, Tianchun Wang, Wei Cheng, Zhuomin Chen, Haifeng Chen, Hua Wei, and Dongsheng Luo. Towards robust fidelity for evaluating explainability of graph neural networks. *International Conference on Learning Representations (ICLR)*, 2024.
- Daniel Zügner, Amir Akbarnejad, and Stephan Günnemann. Adversarial attacks on neural networks for graph data. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2018.